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Contribution of Standardization to Agricultural Development in China

Lingyu Li [1], Chenxia Hu [1], Alan L. Wright [2], Gang Lian [3,*], Lijun Zhou [1] and Jing Yang [1]

[1] College of Economics and Management, China Jiliang University, Hangzhou 310018, China; s22071201017@cjlu.edu.cn (L.L.); huchenxia@cjlu.edu.cn (C.H.); zhlj0606@cjlu.edu.cn (L.Z.); yangjing@cjlu.edu.cn (J.Y.)
[2] Soil and Water Sciences Department, University of Florida-IFAS, Gainesville, FL 32603, USA; alwr@ufl.edu
[3] Zhejiang Key Laboratory of Ecological and Environmental Monitoring, Forewarning and Quality Control, Zhejiang Ecological and Environmental Monitoring Center, Hangzhou 310012, China
* Correspondence: liangang@zjemc.org.cn

Abstract: As a key facilitator of agricultural technology diffusion, the development and implementation of agricultural standards significantly shape future agricultural technology innovation. This study analyzes China's agricultural economic development from 2004 to 2019. It integrates agricultural standard stocks and patent stocks into a Cobb–Douglas production function. Patents serve as a proxy for technological progress, reflecting innovation capacity, while standards represent the institutionalization and diffusion of technological advancements. The analysis focuses on the average annual growth rates of agricultural capital, labor, patents, and standards to clarify their respective contributions to agricultural development. The results show that capital contributes 19.951%, though constrained by inefficiencies. Labor accounts for 38.779% but shows negative elasticity (−0.403%), reflecting the decline of rural labor and the need for mechanization. Patents contribute the most at 42.021%, though limited by weak technology diffusion. Agricultural standards contribute 5.323%, with their impact restricted by adoption barriers. The findings highlight the growing role of technological innovation in agriculture. Strengthening standardization systems and promoting broader adoption of agricultural standards are key to advancing sustainable agricultural development in China.

Keywords: standardization; agriculture; production function; agricultural development; agricultural economics

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1. Introduction

Achieving the Sustainable Development Goals (SDGs) in the coming years is a very important milestone for the sustainability of the planet [1]. There are studies that prove that the SDGs can be achieved through agriculture, which is directly and indirectly linked to all the SDGs [2]. Moreover, standards play an important role in the process of agricultural development. After joining the WTO, China's integration into the global market deepened, highlighting the importance of standardization for economic development. China's national standard GB/T 20000.1-2014 [3] "Standardization Work Guidelines Part 1" defines "standard" as "A document, developed by consensus through standardization activities and in accordance with prescribed procedures, that provides rules, guidelines or characteristics for common use and reuse for a variety of activities or their results" and says that "Standardization is the activity of establishing common and reusable terms for real or potential problems and of preparing, publishing and applying documents in

order to obtain the best possible order within a given scope and to promote co-benefits". China's agricultural standards aim to guarantee the safety of agricultural products, promote sustainable agricultural development, enhance production efficiency, and standardize the entire process of agricultural production (from inputs to processing and distribution). Specifically, agricultural standards mainly involve product standards, technical regulations, input standards, and safety standards.

Standardization influences both macroeconomic and microeconomic areas by improving product quality, optimizing production efficiency, fostering technological innovation, and promoting international trade [4]. This is crucial for advancing China's agricultural economic growth. Agriculture is key to China's economy. Despite becoming a leading global agricultural producer, challenges such as rising consumer demand and climate change persist. To address these, China aims for high-quality agricultural development, transitioning from traditional methods to efficient, sustainable practices. The "Implementation Plan for Standardized Agricultural Production (2022–2025)" seeks to enhance agricultural practices by improving variety, quality, branding, and standardization.

Economic growth theory emphasizes labor, capital, and technical progress as growth drivers [5,6]. This study uses agricultural standards and patents as two indicators of innovation output in agriculture in an economy, arguing that they represent technological progress in agriculture. From there, it comes to assess their impact on China's agricultural development. Many economic scholars contend that technical progress drives innovation activities, which serve as the key driving force of long-term economic growth [7]. Numerous studies have demonstrated the positive growth effect of patents [8,9]. However, the role of standards in economic growth is less studied, especially in agriculture. This paper examines China's agricultural sector from 2004 to 2019, incorporating the stock of standards into the Cobb–Douglas production function alongside agricultural output, capital, labor, and patents. The regression model measures the impact of standardization on agricultural development.

The paper is constructed as follows: Section 2 reviews the literature, Section 3 outlines the methodology and data processing, Section 4 analyzes the results, and Section 5 summarizes conclusions and recommendations.

2. Literature Review

Standards are a set of principles, guidance, or characteristics that have been developed by consensus and adopted by a universally recognized body. Standards serve as a repository of knowledge, promoting innovation and disseminating technologies [10]. The relationship between standards and innovation is well-established, as innovation continuously promotes economic growth. However, despite the significance of standardization, a bibliometric analysis by Heikkilä et al. [11] revealed a surprising lack of attention to the relationship between economic progress and standardization by major scholars in the field of economics. The study found an unexpected result in that no articles were published analyzing this relationship in the Journal of Economic Growth or in the top five economics journals between 1996 and 2018.

Early research on the relationship between standardization and economic growth was mainly conducted at the national level, with studies conducted by national standardization bodies or individual researchers in countries such as the United Kingdom, Germany, and France. Across these studies, a consistent finding emerged: a positive influence of standards in relation to economic development. Jungmittag et al. [12] demonstrated the importance of determining aggregate economic activity using German data from 1961 to 1996. Based on this research, the DIN (German Institute for Standardization) subsequently estimated

economic gains from the standards system would steadily increase at 0.7% to 0.8% of GDP during the study period [13].

Similarly, under the leadership of Miotti [14], the research of AFNOR's (Association Francaise de Normalisation) Marketing and Innovation Department revealed that standardization directly contributed to French economic growth, being responsible for up to 0.81%. The CBoC (Conference Board of Canada) studied the economic influence of standardization and found that standardization accounted for 17% of the country's growth in labor productivity and 9% of growth in output (real GDP) over the period of 1981–2004. The impact of this positive contribution to output growth over time has been substantial. Without growth in standards over the study period, the output (real GDP) in 2004 would have been much lower [15]. Within New Zealand, Stokes et al. [16] highlighted the greater influence of standards on GDP and the wider economy from the impact on total factor productivity (TFP), affecting both labor and capital productivity. Hogan et al. [17] conducted a time series study for the British national standardization body BSI which revealed that the elasticity of standards was 0.11%. Standards contributed 0.7% to GDP growth and 29.4% to total GDP growth. Systematic studies in the Nordic economies also confirmed the fact that standardization significantly increased GDP and labor productivity in all Nordic countries [18]. Collectively, these studies underscore the significant role of standards in driving economic growth across various countries.

Blind and Jungmittag [19] conducted research focusing on the manufacturing sector in the United Kingdom, Germany, Italy, and France from 1990 to 2001. Their analysis, utilizing a Cobb–Douglas production function, revealed technical standards' significant influence on economic development in the 1990s, particularly in less research and experimental development (R&D)-intensive industries. In order to provide insights into the long-term impact of standards on economic development in 11 countries (out of the EU-15), based on the previous studies, Blind et al. [20] first employed panel cointegration techniques. The results of this study were quite similar to those of the 2008 study in which findings reaffirmed that the stock of international and European standards drives economic growth in the EU.

Despite the extensive research in various countries, only a few studies have focused on China. In addition to their impact on national economic growth, standards have a notable influence on sustainable industrial development, particularly in technology-intensive sectors. In recent years, studies in China have provided varied perspectives on this topic. For instance, Bi and Hu [21] examined the impact of standardization on the electronic communications industry in Shandong Province from 1995 to 2012, concluding that the stock of standards positively contributes to industrial output.

In the field of agriculture, the lack of research on agricultural standards and sustainable development of the agricultural economy is evident. Current scholars primarily focus on the impacts of factors such as climate change, technological progress, and finance on agricultural production and economic development. Arora [22] highlights the detrimental effects of climate change, which include enhanced desertification, nutrient-deficient soils, severe drought conditions, and floods. All of these climate-related disasters would have a significant impact on agricultural productivity. Additionally, research by Chen et al. [23] revealed an inverted U-shaped relationship of crop yields with climate change, further emphasizing the complexities of this issue. Moreover, a systematic literature review conducted by Liu et al. [24], emphasized the potential of information and communications technologies (ICTs) and blockchain technologies (BTs) to revolutionize agricultural development. The ICTs are mainly used to enhance yields, while BTs concentrate on improving transparency and transactional efficiency across the agricultural sector through visibility, traceability, and automation. Additionally, the utilization of the Internet of Things (IoT) has

had a significant impact on precision agriculture. Through IoT, farmers manage their farms more effectively, reducing waste of labor and other resources, thus achieving significant increases in efficiency and profitability [25].

Agricultural standards have far-reaching environmental impacts on sustainable development by influencing agricultural production behaviors, guiding technological upgrades, and reshaping ecological relationships. Tayleur et al. examine the potential of voluntary sustainability standards (VSSs) to support biodiversity and agricultural sustainability. Their findings indicate that although all the standards reviewed include provisions for biodiversity protection, only two explicitly prohibit deforestation. Despite certified cropland expanding by 11% annually, it accounts for just 1.1% of global cropland [26]. Schouten and Bitzer explore the emergence of Southern standards in global agricultural value chains, particularly in palm oil, soy, and fruit production. These standards target different stakeholder groups and draw on alternative sources of legitimacy, creating cognitive and moral distance from established norms and raising important implications for sustainability governance [27]. Amekawa et al. assessed public good agricultural practice (GAP) standards in Thailand by comparing 41 certified and 90 non-certified cabbage farmers in Chiang Mai Province. These standards target different stakeholder groups and draw on alternative sources of legitimacy, creating cognitive and moral distance from established norms and raising important implications for sustainability governance [28]. Traldi reviewed 45 studies on 13 major VSSs and highlighted a mismatch between certified crops and research focus: coffee and Fairtrade certification are over-represented, while crops such as cotton and palm oil are under-studied. Most studies emphasize economic outcomes, with limited attention to social and environmental impacts. Overall, 51% of the cases report positive outcomes, though results are often highly context-dependent [29].

3. Research Methodology and Data Processing

3.1. Research Methodology

The Cobb–Douglas production function is a fundamental tool in economic theory, offering insights into production behavior, profit maximization, and cost structure across various industries [30]. Building upon the quantitative method established by Jungmittag et al. [12], this study utilized a simple Cobb–Douglas production function to investigate agricultural standardization's contribution to China's agricultural development.

To clarify the mechanism by which agricultural patents and standards influence agricultural production, this study introduces them as distinct inputs in the production function. Both represent key dimensions of technological progress, yet they differ in terms of accessibility and application. Agricultural patents protect technological innovations through legal rights, granting exclusive use to inventors for a limited time. In contrast, agricultural standards are public documents, typically developed through consensus among stakeholders and approved by regulatory bodies, facilitating widespread access and adoption [31].

Patents and standards reach the agricultural sector through different pathways. Patents often lead to the development and commercialization of new agricultural technologies and products, such as water-saving irrigation systems, pest-resistant genetically modified seeds, and intelligent farming machinery. These innovations are transferred to farmers via technology licensing, enterprise-led commercialization, and government-supported extension services. Their application directly enhances agricultural productivity by increasing efficiency, reducing input costs, and improving crop yields. Agricultural standards, on the other hand, focus on the diffusion and standardized application of advanced technologies and best practices. They promote consistent production processes, ensuring quality control, safety compliance, and environmental sustainability. By reducing information asymmetry and transaction costs, standards facilitate large-scale technology

dissemination across regions and farming entities. This improves resource allocation efficiency, lowers production risks, and reduces the overall cost of agricultural production.

Although technological progress in agriculture encompasses many aspects—including improvements in production techniques, cultivation methods, pest control, fertilization programs, and farm management practices—these factors are often difficult to quantify and incorporate into empirical research. This study does not explicitly quantify the impact of changes in fertilization or crop protection intensity, which could influence productivity. Future research could explore these effects to provide more comprehensive policy insights. Patents and standards are widely used as measurable and accessible indicators of technological progress. In this study, patents reflect the sector's capacity for innovation, while standards represent its ability to diffuse and institutionalize technological advancements. Together, they play a crucial role in optimizing resource use, enhancing production efficiency, and promoting sustainable agricultural development.

The Cobb–Douglas production model for agricultural standardization and agricultural development is assumed as follows:

$$AC_t = AA_t \times AK_t^{\alpha} \times AL_t^{\beta} \times APAT_t^{\gamma} \times ASTD_t^{\theta} \tag{1}$$

In Equation (1), t represents time, where $t = 1, 2, \cdots, n$. Among all the variables, "A" simply represents agriculture. AC_t represents the output of the agriculture sector, including agriculture, forestry, animal husbandry, and fisheries, in period t. The value of agricultural sector output serves as a key indicator of the sector's overall development over the study period. In this study, AA_t is treated as a constant, representing the level of change in agricultural technology during period t. AK_t represents agricultural capital input, AL_t represents agricultural labor input, $ASTD_t$ represents the stock of agricultural standards, and $APAT_t$ represents the stock of agricultural patents at time t. The superscripts α, β, γ, and θ represent the respective production elasticities.

To address heteroskedasticity, we take the natural logarithm of each side of Equation (1) simultaneously and add the regression error term μ_t to the right side. The new regression model is shown in Equation (2):

$$\ln AC_t = \ln AA_t + \alpha \ln AK_t + \beta \ln AL_t + \gamma \ln APAT_t + \theta \ln ASTD_t + \mu_t \tag{2}$$

In Equation (2), the production elasticity coefficient θ represents the impact of agricultural standardization on agricultural development. Through parameter estimation, the value of θ can be obtained. If $\theta > 0$, it indicates that agricultural standardization has a positive impact on agricultural development. Conversely, if $\theta < 0$, it suggests that agricultural standardization hurts agricultural development. If $\theta = 0$, it indicates that agricultural standardization does not affect agricultural development. Furthermore, the magnitude of $|\theta|$ reflects the strength of the effect.

3.2. Variable Selection and Data Processing

(1) Indicator of Agricultural Output (AC_t)

The output of agriculture is typically measured using the primary sector value added. The primary sector encompasses agriculture, forestry, animal husbandry, and fishery in China's industrial classification. Therefore, the agricultural added value of these industries is chosen as the measure of agricultural output. However, using calendar year value added presents challenges due to associated data-processing issues, particularly the influence of inflation. The presence of inflation would lead to a biased determination of agricultural output. This ultimately affects the accuracy of the model estimates.

The nominal value added of the primary sector is transformed into constant prices. This transformation can mitigate the effects of inflation and ensures comparability across calendar years. This transformation process involves several steps. Firstly, time-series data of the nominal value added of the primary sector (in billions of yuan) and value-added index (prior year basis, set to 100) for all years from 2003 to 2019 are obtained from the *China Statistical Yearbook*. Secondly, the base period, which is set to the year 2003, serves as the reference point for the transformation. The base period price is used as the standard for the transformation. Consequently, the value added of the primary sector at constant prices for the years 2004 to 2019 is calculated using the following formula:

$$AC_t = AC_0 \times \prod \frac{r_t}{100} \; t = 1,2,\cdots,n \tag{3}$$

In Equation (3), AC_0 represents the nominal value added of the primary sector in the base period, which is equivalent to the value added at $t = 0$. r_t denotes the index of value added of the primary sector in period t. AC_t represents the value added of agriculture, forestry, animal husbandry, and fishery in period t at constant prices.

(2) Indicator of Agricultural Capital Input (AK_t)

In recent research, capital inputs are usually measured by capital stocks. Initially, agricultural capital stock was chosen as the measure of agricultural capital input in this study. However, upon investigation, it was discovered that statistical data on the capital stock by sector of China's national economy are not readily available. Given the absence of direct statistical data, calculations using the perpetual inventory method were deemed necessary. This approach considers the accumulation of fixed asset investment flows in previous periods as the current capital stock. This stock is extrapolated from data relating to capital formation, investment in fixed assets, and capital depreciation rates.

The data on investment in fixed assets (excluding farm households) in the primary sector which were applied in perpetual inventory method were obtained from the *China Statistical Yearbook*. Similar to the method for calculating agricultural output, the perpetual inventory method requires the selection of a base period. Subsequently, the constant value of fixed asset investment (excluding farm households) and a depreciation rate were used to calculate the capital stock. The detailed calculation method is shown as follows:

$$AK_t = AK_{t-1}(1-\gamma) + AI_t \;\; t = 1,2,\cdots,n \tag{4}$$

The effective stock of capital in the base period is estimated as follows:

$$AK_0 = \frac{AI_0}{\varphi + \gamma} \tag{5}$$

In Equation (4), AK_t represents the capital stock in period t. AK_{t-1} is the capital stock in period $t-1$. AI_t denotes the investment in fixed assets in period t, converted to constant prices. γ represents the depreciation rate of capital. In previous research focusing on China's economic development, it has been generally accepted that the capital depreciation rate is a fixed value. Hall and Jones [32] conducted a study involving 127 countries and found that a depreciation rate of 6% aligns better with reality and yields more accurate results. Therefore, this study adopts a depreciation rate of 6%.

In Equation (5), AK_0 represents the effective stock of capital in the base period, while AI_0 denotes the amount of investment in fixed assets in the base period. φ represents the arithmetic average of the growth rate of annual investment in new fixed assets from year 0 to year t. γ represents the depreciation rate of the capital stock.

(3) Indicator of Agricultural Labor Input (AL_t)

This paper takes employment in the primary sector from 2004 to 2019 as a measure of agricultural labor input. The data on employment in the primary sector were directly obtained from the official data query platform "National Data". This platform, under the purview of the National Bureau of Statistics of China, is tasked with organizing, leading, and coordinating national statistical work. The data provided by this platform are characterized by their truthfulness, accuracy, and timeliness.

(4) Indicator of Agricultural Patent Input ($APAT_t$)

Patents serve as a significant indicator of technological innovation [33,34]. The indicator of agricultural patent input used in the study is represented by the stock of active patents in agriculture over time. This stock is derived from the cumulative number of patents disclosed (published) in period t minus the number of patents that have lapsed in the current period, yielding the effective stock of patents in period t. The study encompassed four categories of patent: invention grant patents, utility model patents, invention application patents, and design patents. However, obtaining comprehensive patent data poses challenges, particularly regarding the amounts of patents disclosed and expired in a calendar year. Only the amount of patents for inventions granted in a calendar year can be obtained from publicly available statistical yearbooks and official data. To overcome these challenges, we decided to use the "Qizhidao" (Enterprise Knowledge) patent search platform operated by a Chinese company. By using the international IPC classification number "A01 (Agriculture)" as the search term, we accessed China's domestic patent database and retrieved agricultural patents. Through further statistical analysis, we obtained the stock of patents in agriculture.

Preliminary analysis of the types of agricultural patents obtained revealed notable characteristics. Among all valid patents in agriculture in China, agricultural machinery patents accounted for approximately 25%, agricultural pesticides and other agricultural chemical products patents accounted for about 20%, and agricultural breeding and cultivation patents accounted for roughly 39%. These three categories of agricultural patents collectively comprised approximately 84% of all agricultural patents, indicating a concentration of patents across specific categories.

(5) Indicator of Agricultural Standard Input ($ASTD_t$)

In China, standards are typically categorized into international standards and domestic standards, with domestic standards further subdivided into five categories: national standards, industry standards, local standards, group standards, and enterprise standards. Among these, national and industry standards hold broader applicability and are the primary focus of research on Chinese domestic standards. The indicator for agricultural standard input can be quantified by the stock of effective standards within a calendar year. This calculation involves subtracting the number of standards repealed or replaced from the cumulative number of standards implemented during the period. The stock of effective agricultural standards only contained national and industry standards. The National Standard Information Public Service Platform is the primary platform for accessing information about Chinese standards. This platform features clear categorization, unique numbering, and detailed publication and revocation dates for standards, ensuring accurate information retrieval. Access to agricultural standards is facilitated through specific selection criteria. Utilizing the International Standard Classification Number (ICS) "65. Agriculture" in the National Standard Catalog allows access to information about national agricultural standards, while information on agricultural industry standards is obtained through China's industry standard classification number, which is "NY Agriculture".

A preliminary examination of China's national agricultural standards reveals notable characteristics. Agricultural and forestry standards constitute approximately 38% of the total, followed by agricultural machinery, tools, and equipment standards at around 21%. Standards for pesticides and other agrochemicals account for about 15%, while feed standards represent approximately 10%, and fertilizer standards comprise only 3%. This distribution highlights an imbalance in agricultural standards, with standards for agriculture and forestry and agricultural machinery, tools, and equipment comprising a significant proportion, almost 60%, of all national agricultural standards. In contrast, other types of standards are relatively scarce.

The initial data collected for this paper are shown in Table 1. Data for five variables from 2004–2019 are included, and there are no missing values in the data.

Table 1. Initial data of the study.

Year	AA_t	AK_t	AL_t	$APAT_t$	$ASTD_t$
2004	18,475.58	2438.42	34,830	395	1797
2005	19,436.31	2847.87	33,442	707	2003
2006	20,408.12	3241.10	31,941	1123	2577
2007	21,163.22	3632.73	30,731	1715	3101
2008	22,306.04	4053.02	29,923	2604	3561
2009	23,242.89	4432.78	28,890	3959	4062
2010	24,242.34	4812.18	27,931	5951	4402
2011	25,284.76	5211.41	26,472	9057	4767
2012	26,422.57	5594.26	25,535	14,063	4770
2013	27,479.47	5956.21	23,838	23,640	4958
2014	28,633.61	6299.95	22,372	37,901	5166
2015	29,778.96	6610.43	21,418	58,050	5501
2016	30,821.22	6898.16	20,908	82,840	5856
2017	32,084.89	7208.31	20,295	115,698	6161
2018	33,239.94	7538.96	19,515	167,285	6570
2019	34,303.62	7869.60	18,652	225,816	6756

4. Results and Discussion

4.1. The Relationship Between Agricultural Standardization and Agricultural Development

To further investigate the influence of agricultural standardization on agricultural development in China, this study conducted the previously established equations using data collected from 2004 to 2019. Before incorporating the above-presented variables, correlation and multicollinearity tests were conducted using SPSS 25.0. Table 2 presents the matrix of correlation coefficients between variables. The results revealed that $\ln AK_t$ was negatively correlated with $\ln AL_t$ and positively correlated with $\ln APAT_t$ and $\ln ASTD_t$. Additionally, $\ln AL_t$ shown negative correlations with $\ln APAT_t$ and $\ln ASTD_t$, while $\ln APAT_t$ exhibited a positive correlation with $\ln ASTD_t$. Moreover, the variance inflation factor (VIF) values for variables exceeded 10 during the multicollinearity test, indicating a significant issue of multicollinearity among the variables.

Table 2. Matrix of correlation coefficients for the independent variables.

Variables	$lnAK_t$	$lnAL_t$	$lnAPAT_t$	$lnASTD_t$
$\ln AK_t$	1.000	−0.972	0.984	0.989
$\ln AL_t$	−0.972	1.000	−0.998	−0.934
$\ln APAT_t$	0.984	−0.998	1.000	0.954
$\ln ASTD_t$	0.989	−0.934	0.954	1.000

The employment of principal component analysis (PCA) can address the issue of multicollinearity among variables. Before conducting PCA, it is necessary to standardize the data and perform related tests. Firstly, all variables were standardized. Data standardization ensures that unit and other differences in the data do not influence the results. Both dependent and independent variables were standardized using Z-scores, computed based on the mean and standard deviation of each variable (Table 3). Subsequently, the standardized dependent variable was denoted as zlnACt, while the standardized independent variables were denoted as $zlnAK_t$, $zlnAL_t$, $zlnAPAT_t$, and $zlnASTD_t$.

Table 3. Means and standard deviations of the independent and dependent variables.

	$lnAC_t$	$lnAK_t$	$lnAL_t$	$lnAPAT_t$	$lnASTD_t$
Mean	10.151	8.516	10.148	9.340	8.343
Standard Deviation	0.197	0.362	0.203	2.031	0.407

To validate the standardized variables, we further conducted the Kaiser–Meyer–Olkin (KMO) and Bartlett's spherical tests. These tests assess both content and structural validity. If the Kaiser–Meyer–Olkin value is higher than 0.6 and the *p*-value of Bartlett's test is lower than 0.05, then the data is suitable for PCA (principal component analysis). In this study, the KMO value is 0.746, and the *p*-value for Bartlett's spherical test is less than 0.01. These results demonstrate that the data are suitable for principal component extraction. As shown in Table 4, the cumulative contribution rate of the principal components Z_1 and Z_2 amounts to 99.92%. These two new variables effectively represent the majority of the information from the original variables. Combining the results from Tables 4 and 5, the expressions of Z_1 and Z_2 are derived as follows:

$$Z_1 = -0.400zlnAK_t - 1.581zlnAL_t + 1.054zlnAPAT_t - 1.521zlnASTD_t \quad (6)$$

$$Z_2 = 0.780zlnAK_t + 1.273zlnAL_t - 0.724zlnAPAT_t + 1.934zlnASTD_t \quad (7)$$

Table 4. Total variance explanation.

Factors	Initial Eigenvalues		
	Total	Percentage of Variance	Accumulated %
1	3.916	97.891	97.891
2	0.081	2.029	99.920
3	0.003	0.067	99.988
4	0.001	0.012	100

Table 5. Matrix of component score coefficients.

Standardized Independent Variable	Factors	
	1	2
$zlnAK_t$	−0.400	0.780
$zlnAL_t$	−1.581	1.273
$zlnAPAT_t$	1.054	−0.724
$zlnASTD_t$	−1.521	1.934

After extracting the principal components, this paper uses the ADF unit root test to test the time-series data of the two factors and determines whether each variable is smooth or not according to the ADF statistics. As can be seen in Table 6, Z_1 and Z_2 do not reject

the hypothesis of the existence of a unit root, indicating that this set of time-series data is non-stationary. However, after first-order differencing the series, the two variables reject the hypothesis of the existence of a unit root at the 10% significance level. This time series becomes smooth after the first-order differencing and is an I(1) single-integrated series, which satisfies the conditions for the cointegration test.

Table 6. ADF test results.

Factors	t-Statistic	Prob.	Threshold Value		
			1% Level	5% Level	10% Level
Z_1	−0.389	0.884	−4.058	−3.120	−2.701
$d(Z_1)$	−2.824	0.082			
Z_2	−2.404	0.141	−4.473	−3.290	−2.772
$d(Z_2)$	−3.235	0.018			

The result of the Durbin–Watson test was 1.067 which showed that there was a slight serial correlation between the data, so the Newey–West standard error method was used for estimation. Regression estimation was performed on $zlnAC_t$ and Z_1 and Z_2. As shown in Table 7, the coefficients of both Z_1 and Z_2 are significant at the 1% level, and the adjusted R^2 is 0.999. These results demonstrate that the overall fit of the model is high and that the estimates are accurate. The White test value was 0.538, indicating that there was no heteroskedasticity problem. The new regression equation, combined with the information in Table 7, can be represented as follows:

$$zlnAC_t = 0.773Z_1 + 0.633Z_2 + c \tag{8}$$

Table 7. Estimated results.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
Z_1	0.773	0.012	63.699	0.000
Z_2	0.633	0.008	76.656	0.000
c	0.000	0.012	0.000	1.000

The purpose of the cointegration test is to test whether there is a stable relationship among the time series in the long run. The residuals of the above regression equation are subjected to the unit root test, and the type of test without a trend and intercept term is used to test whether there is a stable and equilibrium relationship between the variables in the long run; the results are as follows.

From the test results in Table 8, it can be seen that the ADF value of the residuals is less than the critical value of 5% of the significance level; that is, the residual series passes the smoothness test at 5% of the significance level, which indicates that there is a cointegration relationship between the two variables, and there exists a stable equilibrium relationship in the long run.

Table 8. ADF test for residuals.

t-Statistic	Prob.	Threshold Value		
		1% Level	5% Level	10% Level
−3.764	0.017	−4.058	−3.120	−2.701

After substituting Equations (6) and (7) into Equation (8), we obtain Equations (9) and (10):

$$\begin{aligned} zlnAC_t = (0.773 \times (-0.400) + 0.633 \times 0.780)zlnAK_t + (0.773 \times (-1.581) + 0.633 \times 1.273)zlnAL_t \\ +(0.773 \times 1.054 - 0.633 \times 0.724)zlnAPAT_t + (0.773 \times (-1.521) + 0.633 \times 1.934)zlnASTD_t \end{aligned} \quad (9)$$

$$zlnAC_t = 0.185zlnAK_t - 0.416zlnAL_t + 0.355zlnAPAT_t + 0.049zlnASTD_t \quad (10)$$

To calculate the coefficients α, β, γ, and θ in Equation (2), we use the coefficients from Equation (10) and the means and standard deviations in Table 3. The specific calculation process is shown as follows:

$$\begin{aligned} \alpha &= 0.197 \times 0.185/0.362 = 0.101 \\ \beta &= 0.197 \times (-0.416)/0.203 = -0.403 \\ \gamma &= 0.197 \times 0.355/2.031 = 0.034 \\ \theta &= 0.197 \times 0.049/0.407 = 0.024 \end{aligned}$$

The calculation of the coefficients α, β, γ, and θ enables the further calculation of the constant term ln AA_t of the original production function.

$$\ln AA_t = 10.151 - (0.101 \times 8.516 - 0.403 \times 10.148 + 0.034 \times 9.340 + 0.024 \times 8.343) = 12.861$$

The relationship between agricultural standardization and agricultural development, incorporating the constant term $lnAA_t$, is as follows:

$$\ln AC_t = 12.861 + 0.101\ln AK_t - 0.403\ln AL_t + 0.034\ln APAT_t + 0.024\ln ASTD_t \quad (11)$$

4.2. Analysis of Factor Elasticities

The analysis based on Equation (11) reveals significant insights into the relationship between input factors and agricultural output in China. Specifically, the elasticity coefficients indicate the marginal contribution of each input to agricultural production, following the interpretation commonly applied in Cobb–Douglas production functions.

The capital input elasticity is approximately 0.101, implying that a 1% increase in capital investment is associated with a 0.101% increase in agricultural output, holding other factors constant. This highlights the positive and significant role of capital in boosting agricultural productivity. Conversely, the labor input exhibits a negative elasticity of −0.403, suggesting that a 1% increase in labor input corresponds to a 0.403% decrease in agricultural output. This inverse relationship likely reflects a structural transformation in China's agricultural sector: as traditional, labor-intensive practices persist, they constrain productivity improvements. This phenomenon aligns with the shift emphasized at the 2022 Central Rural Work Conference, which advocates transitioning from labor-intensive agriculture towards capital-intensive and technology-driven development. It is also possible that this phenomenon occurs because of factors such as low productivity, inefficiency, or technological gaps in current agricultural labor inputs.

Furthermore, technological factors, represented by the elasticity of patents (0.034) and standards (0.024), contribute positively to agricultural output. A 1% increase in the stocks of patents and agricultural standards leads to 0.034% and 0.024% increases in agricultural output, respectively. Although the elasticity of patents (0.034) and standards (0.024) indicates a relatively small marginal impact on agricultural output, these factors play a crucial strategic role in promoting long-term sustainable development. Technological innovations embodied in patents and standards drive productivity gains through improved mechanization, efficiency, and quality control. Their effects, while gradual, are cumulative and transformative, supporting the transition to a modern, technology-intensive agricultural system.

Since 2008, there has been a drastic increase in the total capital input in China's agricultural sector [35]. This surge can be attributed to rising labor costs, prompting more and more household farms to opt for capital over labor [36,37]. As more household farms shift towards capital-intensive production modes, the positive elasticity of the capital input becomes more impactful. However, with the constraint of technology [38], de-specialized production remains prevalent in Chinese agriculture [39]. Small landholders continue to dominate agricultural production. This reliance on small-scale farming, characterized by non-mechanized production methods, significantly lowers agricultural productivity compared to professional mechanized production.

In conclusion, elasticity analyses suggest that the substitution of capital for labor and technological progress are key drivers of agricultural production growth in China. However, overcoming persistent challenges, such as the prevalence of small landowners and the limited adoption of mechanized and standardized production, remains critical to maximizing agricultural productivity and ensuring long-term sustainability.

4.3. Analysis of Factor Growth Rates

Before analyzing the cumulative contribution of the input factors to agricultural development, it is essential to examine the growth rate of each factor. The growth rate of each factor is calculated with the help of the raw data presented in Table 1. From 2004 to 2019, China's agricultural output experienced an average annual growth rate of 4.344%. In contrast, the average annual growth rates of capital, labor, the stock of patents, and standards were 9.000%, −4.054%, 52.026%, and 10.062%, respectively (see Figure 1). Two distinct features emerge from this analysis. Firstly, the average annual growth rate of the stock of patents far exceeds that of capital and the stock of standards. Secondly, the average annual growth rate of labor exhibits a negative value.

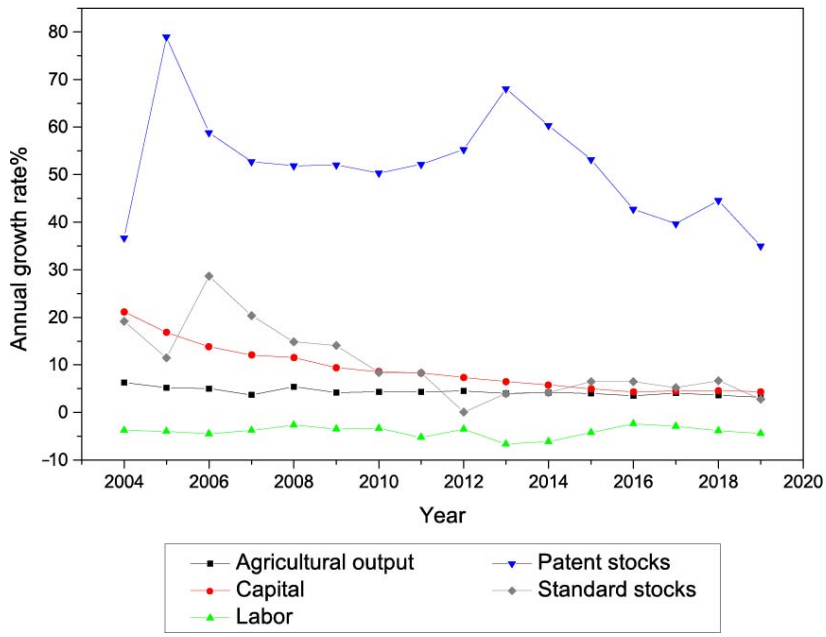

Figure 1. Annual growth rates of factors, 2004–2019.

Examining the trends in the average annual growth rates of each factor reveals notable differences in their fluctuations. The growth rates of the stocks of patents and standards display pronounced fluctuations, with peaks observed in 2005 and 2013 for patent stocks and in 2006 for standard stocks. Subsequently, both experienced rapid declines post-peak, with the growth rate of patent stocks in 2019 being even lower than that in 2004. The growth

rate of standard stocks has stabilized in recent years after reaching its lowest point in 2012 at 0.063%. In contrast, the trends in the annual growth rates of agricultural output, capital, and labor have remained relatively flat and consistently low. Specifically, agricultural output has maintained an annual growth rate of around 3%, while capital's annual growth rate has steadily declined from 21.185% in 2004 to approximately 4%. Notably, the labor factor consistently exhibits a negative growth rate, reflecting the ongoing decline in China's rural workforce and the aging of its population. Among labor-intensive industries, agriculture has been one of the sectors most affected by the aging of the population [40,41]. The factor growth rates analysis underscores the dynamic nature of input factor growth rates in China's agricultural sector, and the results highlight the challenges posed by demographic shifts and the need for targeted policy interventions to sustain agricultural productivity and development.

4.4. Analysis of Cumulative Contribution of Factors

Various input factors have made different contributions to the sustainable development of Chinese agriculture in recent years. To explore the specific factors driving this growth, this study calculates the contribution rates of key input variables based on their elasticity coefficients and growth rates. The calculation follows Equation (12), which measures each factor's contribution relative to the average annual agricultural GDP growth rate:

$$\text{Factor contribution rate} = \text{Annual growth rate of factor} \times \text{elasticity} / \text{Average annual GDP growth rate} \quad (12)$$

The results derived from Equation (11) reveal considerable differences in the contribution rates of these factors. Specifically, patent stocks and labor inputs exhibit the highest average annual contribution rates, at 42.021% and 38.779%, respectively, while capital and standard stocks show lower contributions of 19.951% and 5.323%, respectively.

Further examination of cumulative contribution rates depicted distinct trends among the input factors. Figure 2 represents the trend of the cumulative contribution rates. While labor's cumulative contribution fluctuated slightly over the years, capital, patent stocks, and standard stocks showed a steady upward trend. Particularly, patent stocks have surpassed labor in cumulative contribution since 2005; this trend underscores the growing importance of technological progress, represented by the growth of patent stocks, as a driver of agricultural development. Of particular interest is the widening gap between the cumulative contribution rates of capital and labor, especially after 2008, suggesting a divergence in their respective contributions to agricultural development. However, it is essential to clarify that the 42.021% contribution attributed to patent stocks does not imply that patents alone account for this share of agricultural production growth. Rather, patent stocks serve as a proxy for overall technological progress, which manifests through the adoption and diffusion of modern technologies, such as mechanization, improved crop varieties, precision agriculture, and standardized production processes. In this context, technological progress enhances capital efficiency and labor productivity, enabling higher agricultural output. Thus, technological progress, capital investment, and labor inputs are interdependent, jointly driving agricultural modernization.

The widening gap between the cumulative contribution rates of capital and labor in China's agricultural sector contradicts the country's ambition to transition to a capital-intensive agriculture industry. Although capital inputs have increased, particularly after 2008, the lower-than-expected contribution rate of capital (19.951%) suggests inefficiencies in the deployment and utilization of financial resources. This discrepancy may stem from several factors, including the unreasonable structure of fiscal inputs, the misalignment between agricultural investment and fiscal policy objectives, and imperfect coordination mechanisms among relevant stakeholders [42]. These issues may result in significant

resource wastage and hinder the optimal utilization of available funds for agricultural sustainable development initiatives.

At the same time, labor continues to be the primary driver of China's agricultural development, contributing 38.779% of agricultural growth. However, labor's role is changing. With the expected reduction in rural populations in the future, the number of individuals directly involved in agricultural production and management is expected to decrease. The aging of farmers and persistently low agricultural productivity, coupled with the demographic shift, exacerbates challenges such as the "hollowing out of the countryside" and the prevalence of "aging farmers" [43,44]. The reliance on labor-intensive production persists, but productivity per labor unit is relatively low. For instance, when comparing China to the United States, China's agricultural value added is approximately 5.8 times that of the United States, yet total agricultural employment in China is about 95 times higher. This imbalance underscores the urgent need for productivity enhancements, through mechanization and technology adoption, to ensure sustainability as the labor force declines.

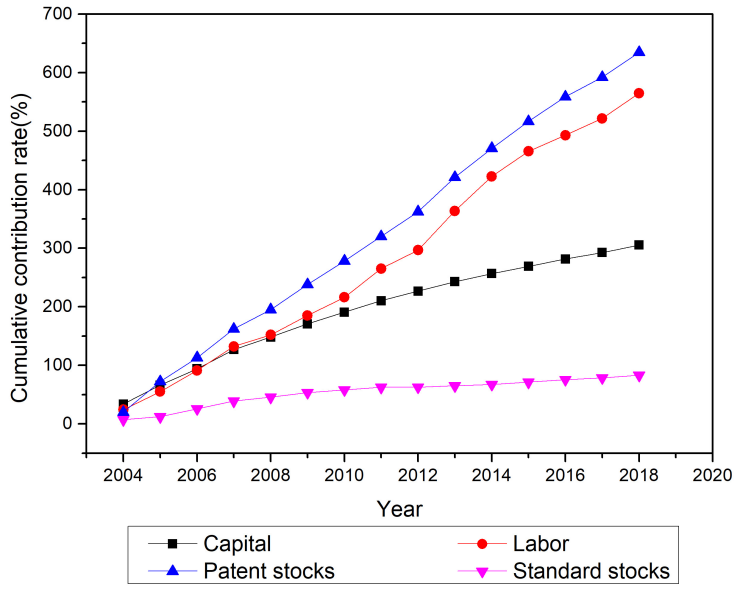

Figure 2. Cumulative contribution rates of factors to agricultural output, 2004–2019.

The high contribution rate of patent stocks (42.021%) reflects the growing role of technological innovation in driving agricultural output, but there are significant challenges in technology diffusion and commercialization. Despite increased patent filings, the application and practical impact of agricultural patents remain limited. This is mainly due to weak independent innovation in agricultural machinery and breeding technology, lagging digital agricultural infrastructure, and limited precision agriculture and information systems. The driving force of agricultural patents for technology innovation remains suboptimal and requires enhancement. Patented technologies need to be translated into production tools, such as mechanized equipment, improved seeds, and smart farming systems [45,46]. Furthermore, complementary investments in capital and skill development and targeted policy support are crucial to fully realize the benefits of these innovations.

The cumulative contribution rate of standard stocks (5.323%) remains low. Standards have an essential effect on economic development by fostering the dissemination of self-innovation knowledge [47]. In China, there are still major difficulties in promoting and adopting standardized agricultural practices. Their impact is constrained by factors such as the lack of agricultural technicians at the grass-roots level, limited government funding aimed at promoting the adoption of standards, and the low willingness of farmers and agribusinesses to adopt new technologies and practices. This situation hampers the

widespread application of national and industry standards in agriculture, thereby limiting the positive impact of standards on agricultural sustainable development.

The findings suggest that technological progress is a key driver of agricultural sustainable development in China, but its impact depends on effective integration with capital investment and labor transformation. However, a series of major agricultural policies introduced in China, such as at the 18th Party Congress in 2012, which proposed integrated urban–rural development as a fundamental way to solve rural problems, and the 19th Party Congress in 2017, which proposed the strategy of "revitalization of the countryside", have failed to play their expected roles. Therefore, it is still necessary to fully unleash the potential of technological progress from various aspects, so that agricultural policies, technological innovation, capital investment, and labor level enhancement can be fully integrated, thus promoting the development of the agricultural industry.

5. Conclusions and Recommendations

This study analyzes the contributions of capital, labor, patents, and standards to China's agricultural development. The results show that capital investment positively impacts agricultural productivity, though its contribution (19.951%) remains limited due to inefficiencies in resource allocation and utilization. The labor input, despite its high contribution rate (38.779%), exhibits a negative elasticity (−0.403%), reflecting the challenges of labor-intensive farming and rural population decline. This underscores the urgent need for mechanization and labor transformation. Technological progress, represented by patents and standards, is increasingly important. Patent stocks contribute the most (42.021%) to agricultural growth, though their impact is hindered by weak technology diffusion and commercialization. Standards show a limited contribution (5.323%) due to barriers in adoption and implementation.

There are still some problems in the existing agricultural standardization system. The first is that there is fragmentation and duplication among standards, with overlapping content and even conflict between national standards (GB), industry standards (NY), and local standards (DB). Standards in emerging areas are lagging behind, and standards across the industry chain are disconnected. Secondly, there is uneven implementation of standards and insufficient testing capacity and social credibility. Finally, the implementation cost of green standards is high, and there is an imbalance between standards and biodiversity conservation. We need to recognize the importance of agricultural standardization and make efforts to improve the deficiencies in the agricultural standardization system. Based on the findings of this study, we offer the following targeted policy recommendations to enhance China's agricultural standardization system and promote sustainable agricultural development.

This study provides empirical evidence that both agricultural standards and patents significantly contribute to agricultural productivity. Unlike previous studies that focus mainly on capital and labor, our research highlights standardization as a key institutional driver of technological progress and efficiency improvements. The following measures are recommended:

(1) Improve the Standardization System and Its Implementation

Streamline and integrate national (GB), industry (NY), and local (DB) standards to reduce duplication and contradictions. Accelerate standard development in emerging fields such as smart agriculture and green production. At the same time, enhance implementation by improving testing capacity, certification systems, and technical support services. Provide incentives to encourage widespread adoption, particularly for green and biodiversity-friendly standards.

(2) Optimize Capital Investment Efficiency

The results indicate that capital remains a key driver of agricultural growth when combined with technological progress. Policymakers should guide capital investment toward mechanization, precision farming, and digital technologies to maximize productivity gains. Financial instruments and subsidies can be tailored to support small and medium-sized farms in adopting advanced technologies that comply with modern standards.

(3) Facilitate Labor Transition and Capacity Building

Given the findings on labor's diminishing marginal returns, it is crucial to support the transition from traditional to modern agricultural practices. This includes offering targeted training programs to enhance farmers' technical skills and digital literacy. Policies should also support the rural workforce's mobility toward modern agribusiness sectors or rural industries that benefit from standardized production processes.

(4) Promote Technological Innovation and the Application of Patented Technologies

Our study demonstrates that patents drive innovation but require better mechanisms for technology transfer and commercialization. Strengthening intellectual property protection, expanding extension services, and facilitating public–private partnerships can enhance the adoption of patented technologies. This will bridge the gap between research outputs and practical applications in the field.

By addressing the institutional weaknesses in the standardization system and promoting the integration of standards with technological innovations, China can unlock the full potential of agricultural productivity and sustainability. This study provides novel empirical evidence on the complementary role of patents and standards in agricultural development, offering actionable insights for policymakers to formulate targeted interventions. The implementation of these recommendations will contribute to achieving China's goals of sustainable agriculture, rural revitalization, and high-quality economic development.

6. Limitations and Future Research

First, although agricultural standards and patents are used as proxy variables for technological progress in this study, it is difficult to accurately quantify their specific impact on agricultural production outcomes. The lack of detailed data on the actual implementation and utilization efficiency of patents and standards may affect the precision of the empirical results. Future research could start with micro-level data to find data that reflect the effectiveness of actual enforcement of patents and standards to better capture their differential impacts.

Second, there are limitations due to data availability, especially those related to agricultural standards. This study only focuses on the impact of domestic agricultural standards on China's agricultural development. The potential impacts of international standards and cross-border standardization cooperation are not addressed. Expanding data collection to international agricultural standards would provide a more comprehensive perspective for future research.

Third, the development of the agricultural sector is influenced by a variety of complex and dynamic factors beyond the scope of this study. Factors such as environmental conditions, climate change, government policy interventions, market volatility, and rural infrastructure improvements may interact with agricultural standards and patents over time. The current model does not adequately account for these potential interactions. Future research could explore these dynamics through more sophisticated modeling techniques, such as dynamic panel data models or nonlinear estimation methods.

If these limitations can be addressed in future research, it will improve the robustness and applicability of future research on agricultural standardization and its role in promoting sustainable agricultural development.

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Abbreviations

The following abbreviations are used in this manuscript:

DIN	German Institute for Standardization
AFNOR	Association Francaise de Normalisation (French Institute for Standardization)
CBoC	Conference Board of Canada
TFP	Total factor productivity
BSI	British Standards Institution
R&D	Research and experimental development
EU-15	The initial members of the European Union, including Austria, Belgium, Denmark, Finland, France, Germany, Greece, the United Kingdom, Ireland, Italy, the Netherlands, Portugal, Spain, Sweden, and Luxembourg

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